

Integration Wrap-up

Today's Goals

- ◆ You should be able to employ different strategies to evaluate integrals.
- ◆ You should be able to recognize when an integral will be difficult or impossible to evaluate exactly.
- ◆ You should be able to find formulas for functions when given information about their derivatives.

1. Evaluate each of the following integrals. You won't be able to evaluate one of them—which one is it?

(a) $\int \sin(\sin x) \cos x \, dx$

Solution

$u = \sin x$ is a promising substitution because it's the inner function in the composition $\sin(\sin x)$ and $du = \cos x \, dx$ is also in the integral:

$$\begin{aligned} \int \sin(\sin x) \cos x \, dx &= \int \sin u \, du \\ &= -\cos u + C \\ &= \boxed{-\cos(\sin x) + C} \end{aligned}$$

(b) $\int_e^{e^e} \frac{(\ln x)^5}{x} \, dx$

Solution

It looks like $u = \ln x$ is a promising substitution because it's the inner function in the composition $(\ln x)^5$, and we also see $du = \frac{1}{x} \, dx$ in the integral:

$$\begin{aligned} \int_e^{e^e} \frac{(\ln x)^5}{x} \, dx &= \int_1^e u^5 \, du \\ &= \left. \frac{1}{6} u^6 \right|_1^e \\ &= \boxed{\frac{1}{6}(e^6 - 1)} \end{aligned}$$

(c) $\int_e^{e^e} \frac{\ln(x^5)}{x} \, dx$

Solution

An easy first step is to use log rules to rewrite $\ln(x^5) = 5 \ln x$. It then looks like $u = \ln x$ is a

promising substitution because we also see $du = \frac{1}{x} dx$ in the integral:

$$\begin{aligned}\int_e^{e^e} \frac{\ln(x^5)}{x} dx &= \int_e^{e^e} \frac{5 \ln x}{x} dx \\ &= \int_1^e 5u du \\ &= \frac{5}{2} u^2 \Big|_1^e \\ &= \boxed{\frac{5}{2}(e^2 - 1)}\end{aligned}$$

(d) $\int_0^1 \arctan x \, dx$

Solution

This integral requires us to integrate by parts. Let $u = \arctan x$, $dv = 1 \cdot dx$ so that $v = x$, and $du = \frac{1}{1+x^2} dx$. Then

$$\begin{aligned}\int_0^1 \arctan x \, dx &= x \arctan x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} dx \\ &= \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} dx.\end{aligned}$$

To evaluate $\int_0^1 \frac{x}{1+x^2} dx$, we need to use a substitution. Here, $u = 1+x^2$ seems like a good choice, as $du = 2x \, dx$, and we have an $x \, dx$ in the numerator. Thus

$$\int_0^1 \frac{x}{1+x^2} dx = \frac{1}{2} \int_1^2 \frac{1}{u} du = \frac{1}{2} (\ln(2) - \ln(1)) = \frac{1}{2} \ln 2.$$

Putting this all together, we have

$$\int_0^1 \arctan x \, dx = \boxed{\frac{\pi}{4} - \frac{1}{2} \ln 2.}$$

(e) $\int_0^2 e^{-x^2} dx$

Solution

This is the one we can't evaluate! It turns out that e^{-x^2} has no antiderivative we can make out of familiar functions.

(f) $\int_{-1}^1 x^3 e^{-x^2} dx$

Solution

This is an odd function, so the negative area between $x = -1$ and $x = 0$ will exactly cancel out the positive area between $x = 0$ and $x = 1$. Thus $\int_{-1}^1 x^3 e^{-x^2} dx = \boxed{0}$.

(g) $\int \cos \sqrt{x} dx$

Solution

There are different ways to tackle this integral. The easiest is probably to start with substitution, although that may not look promising at first. Let $y = \sqrt{x}$. (Usually when we substitute, we use the letter u , but we also use the letter u when we integrate by parts, so let's use y for the substitution.) Then, $dy = \frac{1}{2}x^{-1/2}dx = \frac{1}{2y}dx$. Therefore, $dx = 2y dy$, and

$$\int \cos \sqrt{x} dx = \int \cos(y) 2y dy = 2 \int y \cos(y) dy \quad (1)$$

Now, we can use integration by parts to evaluate the integral $\int y \cos(y) dy$. We did this on PSet 35, but let's do it again: Set $u = y$ and $\frac{dv}{dy} = \cos(y)$. Then, $\frac{du}{dy} = 1$ and $v = \sin(y)$. Therefore,

$$\int y \cos(y) dy = y \sin(y) - \int \sin(y) dy = y \sin(y) + \cos(y) + C.$$

Remembering that $y = \sqrt{x}$ and going back to 1, we have

$$\int \cos \sqrt{x} dx = \boxed{2(\sqrt{x} \sin \sqrt{x} + \cos \sqrt{x}) + C}$$

(h) $\int_0^1 \frac{2x+3}{x^2+1} dx$

Solution

We can rewrite this integral as a sum of two integrals and work on the two integrals separately.^a

$$\int_0^1 \frac{2x+3}{x^2+1} dx = \int_0^1 \frac{2x}{x^2+1} dx + \int_0^1 \frac{3}{x^2+1} dx \quad (2)$$

The second integral is one you should just recognize:

$$\begin{aligned} \int_0^1 \frac{3}{x^2+1} dx &= 3 \arctan x \Big|_0^1 \\ &= 3 \cdot \frac{\pi}{4} - 3 \cdot 0 \\ &= \frac{3\pi}{4} \end{aligned}$$

The first integral is not one we just recognize, but the substitution $u = x^2 + 1$ looks useful, both because this is the denominator and because we see $du = 2x dx$ in the integral. As x

goes from 0 to 1, $u = x^2 + 1$ goes from $0^2 + 1 = 1$ to $1^2 + 1 = 2$.

$$\begin{aligned}\int_0^1 \frac{2x}{x^2 + 1} dx &= \int_1^2 \frac{1}{u} du \\ &= \ln |u| \Big|_1^2 \\ &= \ln 2 - \ln 1 \\ &= \ln 2\end{aligned}$$

Putting this all together with equation (2), $\int_0^1 \frac{2x + 3}{x^2 + 1} dx = \ln 2 + \frac{3\pi}{4}$.

^aIs it obvious when you look at this problem that this is the right first step? Probably not! But, as you practice more, you'll start to develop your intuition about how to tackle integrals like this.

2. Approximate the integral you weren't able to evaluate in #1. (Use a calculator to obtain a decimal result.) Can you determine the maximum error in your approximation?

Solution

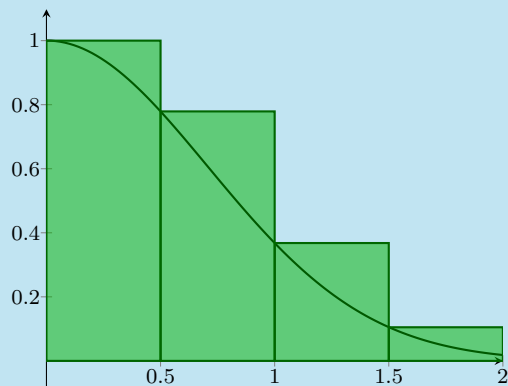
Let's approximate $\int_0^2 e^{-x^2} dx$ using the right-hand sum R_4 .



$$R_4 = \frac{1}{2} \cdot e^{-(0.5)^2} + \frac{1}{2} e^{-(1)^2} + \frac{1}{2} e^{-(1.5)^2} + \frac{1}{2} e^{-(2)^2} \approx \boxed{0.635}$$

Since all of the rectangles lie below the curve, this is an underestimate.

If we instead use the left-hand-sum L_4 , the approximation looks like this:



$$L_4 = \frac{1}{2} \cdot e^{-(0)^2} + \frac{1}{2} e^{-(0.5)^2} + \frac{1}{2} e^{-(1)^2} + \frac{1}{2} e^{-(1.5)^2} \approx 1.126$$

Based on the graph, we can see that the left-hand sum is an overestimate. Therefore, the true value of the integral lies between 0.635 and 1.126. In other words, the maximum error of our original estimate is $1.126 - 0.635 = 0.491$. We could decrease the error in our approximation by using more rectangles.

3. A rocket starts on the ground at rest. At a time of t seconds after it is launched, its acceleration is

$$a(t) = 14e^{t/100} - 10,$$

measured in (meters/second)/second.

- (a) The rocket has a velocity of 0 meters/second at the moment it is launched. Compute the velocity of the rocket t seconds after it is launched.

Solution

Since the acceleration is the derivative of the velocity, we can obtain the velocity of the rocket by taking the antiderivative of the acceleration. So the velocity at t seconds after launch is given by

$$v(t) = 1400e^{t/100} - 10t + c.$$

We know the rocket starts with zero velocity, $v(0) = 0$, implying that $c = -1400$ and we can conclude:

$$v(t) = 1400e^{t/100} - 10t - 1400.$$

- (b) How far does the rocket travel in the first 30 seconds of its movement?

Solution

Let $d(t)$ be the distance traveled by the rocket t seconds after it launches. Then the derivative

of $d(t)$ is $v(t)$, which means that the distance traveled between launch and time t is given by

$$\begin{aligned}d(t) &= \int_0^t 1400e^{s/100} - 10s - 1400 \, ds \\&= 140000e^{s/100} - 5s^2 - 1400s \Big|_0^{30} \\&= 140,000(e^{30/100} - 1) - 5(900) - 1400(30) \\&= \boxed{140,000e^{3/10} - 186,500}\end{aligned}$$